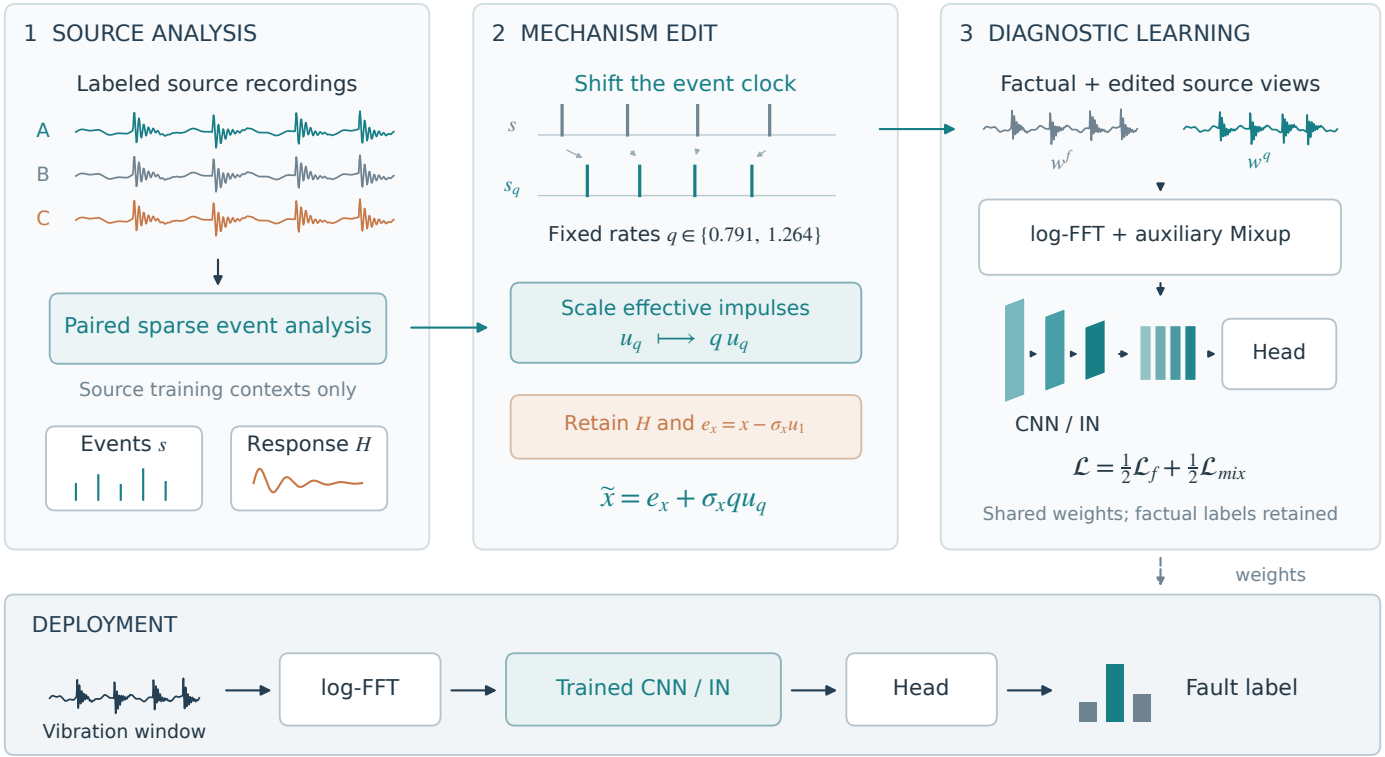
1 SOURCE ANALYSIS
Labeled source recordings
A
B
C
Paired sparse event analysis
Source training contexts only
Events s
Response H
2 MECHANISM EDIT
Shift the event clock
s
s_q
Fixed rates $q \in \{0.791,\ 1.264\}$
Scale effective impulses
$u_q \longmapsto q\,u_q$
Retain H and $e_x = x - \sigma_x u_1$
$\tilde{x} = e_x + \sigma_x q u_q$
3 DIAGNOSTIC LEARNING
Factual + edited source views
w^f
w^q
log-FFT + auxiliary Mixup
Head
CNN / IN
$\mathcal{L} = \frac{1}{2}\mathcal{L}_f + \frac{1}{2}\mathcal{L}_{mix}$
Shared weights; factual labels retained
weights
DEPLOYMENT
Vibration window
log-FFT
Trained CNN / IN
Head
Fault label